UNIT I:
INTRODUCION TO MOBILE COMPUTING

Mobile computing is an umbrella term use to describe the technologies the enables the access to network by people from anywhere and at anytime. As the name implies it is work in motion where geographical boundary and space do not restrict communication through network.
This easy accessibility of information from accessible anywhere and at any time was enabled by technology of cellular communication, wireless LANs, and satellite services Regardless of size, most mobile computers will be equipped with a wireless connection to the fixed part of the network, and, perhaps, to other mobile computers. The resulting computing **environment , which is often referred to as *Mobile Computing* no longer** requires users to maintain a fixed and universally known position in the network and enables almost unrestricted mobility. Mobility and portability will create an entire new class of applications and, possibly, new massive markets combining personal computing and consumer electronics.

Other names for mobile computing
Many other names/overlapping computing paradigms include:
• Pervasive Computing
• Ubiquitous Computing
• Wireless Computing
• Embedded Computing
• Nomadic Computing
• Wireless Sensor Networks
• (Mobile) Ad-Hoc Networks
• Mesh Networks
• Vehicular Networks

Why Mobile Computing?

• It enables anywhere/anytime connectivity
• it brings computer communications to areas without preexisting infrastructure
• It enable mobility
• It enables new applications
• it is an exciting new research area

Evolution of Mobile Computing
Mobile computing can be categorized into seven major categories of focus or eras which are the basis for the technology that is used today in research and design of mobile computing. Each category or section is a different area that was
focused on making mobile computing what it is today. These seven categories are:

i. Portability,
ii. Miniaturization,
iii. Connectivity
iv. Convergence
v. Divergence
vi. Applications

vii. Digital
viii. Ecosystems

Portability

This era focused on reduction of hardware size to enable the creation of computers that could be physically moved around relatively easily

Miniaturization

This era focused in creating new and significantly smaller mobile form factors that allowed the use of personal mobile devices while on the move.

Connectivity

The era of developing devices and applications that allowed users to be online and communicate via wireless data networks while on the move.

Convergence

This is the era of integrating emerging types of digital mobile devices, such as Personal Digital Assistants (PDAs), mobile phones, music players, cameras, games, etc., into hybrid devices (eg. Smart phones)

Divergence

This is opposite approach to interaction design by promoting information appliances with specialized functionality rather than generalized ones.

Applications (Apps)

The latest wave of applications (apps) is about developing matter and substance for use and

consumption on mobile devices, and making access to this fun or functional interactive application content easy and enjoyable.

Digital Ecosystems

The emerging wave of digital ecosystems is about the larger wholes of pervasive and interrelated technologies
that interactive mobile systems are increasingly becoming a part of.

Applications Of Mobile Computing

The ability to keep working on the move is important in many fields of work for effective use of time. The importance of Mobile Computing/Wireless communication has been highlighted in many fields of which a few are described below:

Vehicles

Many wireless communication systems and mobility aware applications are used for following purpose in vehicles:

- Transmission of music, news, road conditions, weather reports, and other broadcast information are received via digital audio broadcasting (DAB) with 1.5Mbit/s.
- For personal communication, a universal mobile telecommunications system (UMTS) phone might be available offering voice and data connectivity with 384kbit/s.

- For remote areas, satellite communication can be used, while the current position of the car is determined via the GPS (Global Positioning System).
- A local ad-hoc network built by cars driving in the same area can be for the fast exchange of information (information such as distance between two vehicles, traffic information, road conditions) in emergency situations or to help each other keep a safe distance.
- Local ad-hoc network with vehicles close by to prevent guidance system, accidents, redundancy.
- Buses, trucks, and trains can transmit maintenance and logistic information to their home base, which helps to improve organization (fleet management), and saves time and money.
- Vehicle data from buses, trucks, trains and high-speed train can be transmitted in advance for maintenance.
- In ad-hoc network, car can comprise personal digital assistants (PDA), laptops, or mobile phones connected with each other using the Bluetooth technology.
- In case of an accident; not only will the airbag be triggered, but the police and ambulance service will be informed via an emergency call to a service provider.

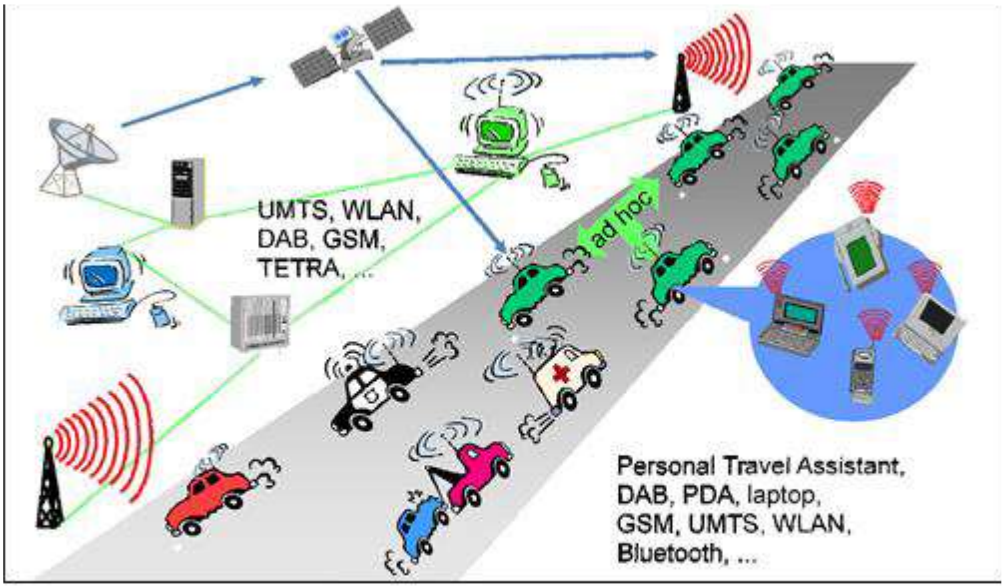


Figure 1.1: Typical Application of Mobile Communication in Road Traffic.

Emergency

Following services can be provided during emergencies:

- Video communication: Responders often need to share vital information. The transmission of real time situations of video could be necessary. A typical scenario includes the transmission of live video footage from a disaster area to the nearest fire department, to the

police station or to the near NGOs etc.

- Push To Talk (PTT): PTT is a technology which allows half duplex communication between two users where switching from voice reception mode to the transmit mode takes place with the use of a dedicated momentary button. It is similar to walkie-talkie.
- Audio/Voice Communication: This communication service provides full duplex audio channels unlike PTT. Public safety communication requires novel full duplex speech transmission services for emergency response.
- Real Time Text Messaging (RTT): Text messaging (RTT) is an effective and quick solution for sending alerts in case of emergencies. Types of text messaging can be email, SMS and instant message.
- An ambulance with a high-quality wireless connection to a hospital can carry vital information about injured persons to the hospital from the scene of the accident. All the necessary steps for this particular type of accident can be prepared and specialists can be consulted for an early diagnosis.
- Wireless networks are the only means of communication in the case of natural disasters such as hurricanes or earthquakes . In the worst cases, only decentralized, wireless ad-hoc network survive.

Business

- A travelling salesman today needs instant access to the company's database: to ensure that files on his or her laptop reflect the current situation, to enable the company to keep track of all activities of their travelling employees, to keep databases consistent etc.
- With wireless access, the laptop can be turned into a true mobile office, but efficient and powerful synchronization mechanisms are needed to ensure data consistency.
- With mobile devices, managers can send critical presentations to major customers. They can also access the latest market share information, communicate with the office about possible new offers and call meetings for discussing responds to the new proposals" Therefore., mobile computers can leverage competitive advantages.

In Office

- Wi-Fi wireless technology saves businesses or companies a considerable amount of money on installations costs.
- There is no need to physically setup wires throughout an office building, warehouse or store.
- Bluetooth is also a wireless technology especially used for short range that acts as a complement to Wi-Fi. It is used to transfer data between computers or cellphones.

Credit Card Verification

- In shops and supermarkets, when customers use credit cards for transactions at Point of Sale (POS) terminals, the intercommunication required between the bank central computer and the POS terminal in order to effect verification of the card usage, can take place quickly and securely over cellular channels using a mobile computer unit. This can speed up the transaction process and relieve congestion at the POS terminals.

Transportation Industries
In transportation industries, GPS technology is used to find efficient routes and tracking vehicles.

Replacement of Wired Network
Wireless network can also be used to replace wired network. Due to economic reasons it is often impossible to wire remote sensors for weather forecasts, earthquake detection, or to provide environmental information, wireless connections via satellite, can help in this situation.
Tradeshows need a highly dynamic infrastructure, since cabling takes a long time and frequently proves to be too inflexible.
Many computers fairs use WLANs as a replacement for cabling. Other cases for wireless networks are computers, sensors, or information displays in historical buildings, where excess cabling may destroy valuable walls or floors.

Location dependent service
It is important for an application to know something about the location because the user might need location information for further activities. Several services that might depend on the actual location can be described below:
Follow-on Services:
Location aware services: To know about what services (e.g. fax, printer, server, phone, printer etc.) exist in the local environment.
Privacy: We can set the privacy like who should get knowledge about the location.
Information Services: We can know about the special offers in the supermarket. Nearest hotel, rooms, cabs etc.

Infotainment: (Entertainment and Education)
- Wireless networks can provide information at any appropriate location.
- Outdoor internet access.
- You may choose a seat for movie, pay via electronic cash, and send this information to a service provider.
- Another growing field of wireless network applications lies in entertainment and games to enable, e.g., ad-hoc gaming networks as soon as people meet to play together.

Limitations of Mobile Computing

- **Resource constraints**: Mobile devices mostly use battery as source of power.
- **Interference:** Radio transmission cannot be protected against interference using shielding, and this result in higher loss rates for transmitted data or higher bit error rates respectively
- **Bandwidth:** Although they are continuously increasing, transmission rates are still very low for wireless devices compared to desktop systems. Researchers look for more efficient communication protocols with low overhead.
- **Dynamic changes in communication environment**: Variations in signal power within a region, thus link delays and connection losses.
- **Network Issues**: Discovery of the connection-service to destination and

connection stability

- **Interoperability issues**: The varying protocol standards
- **Security constraints**: Not only can portable devices be stolen more easily, but the radio interface is also prone to the dangers of eavesdropping. Wireless access must always include encryption, authentication, and other security mechanism that must be effective and easy to use.

UNIT II

MOBILE AND WIRELESS DEVICES

"Mobile device" is a common term used to refer to a wide range of devices that allow people to access data and information from anywhere at any time. These are in different sizes, from those that fit in your pocket, such as cell phones, to tablets, to laptops. Though many are presently available, future is expecting much more devices.

Classification of Mobile devices

There is no precise classification of such devices, by sizes, shape, weight, or computing power. The following list of given examples of mobile and wireless devices graded by increasing performance (CPU, memory, display, input devices, etc.)

Sensor: Wireless device is represented by a sensor transmitting state information, a switch, sensing the office door can be an example. If the door is closed, the switch transmits this information to the mobile phone inside the office which will not accept incoming calls without user interaction; the semantics of a closed door is applied to phone calls.

Embedded Controller: Many applications already contain a simple or sometimes more complex controller. Keyboards, mouse, headsets, washing machines, coffee machines, hair dryers and TV sets are just some examples.

Pager: As a very simple receiver, a pager can only display short text messages, has a tiny display, and cannot send any messages.

Personal Digital Assistant: PDAs typically accompany a user and offer simple versions of office software (calendar, notepad, mail). The typically input device is a pen, with built-in character recognition translating handwriting into characters. Web browsers and many other packages are available for these devices.

Pocket computer: The next steps towards full computers are pocket computers offering tiny keyboards, color displays, and simple versions of programs found on desktop computers (text processing, spreadsheets etc.)

Notebook/laptop: Laptops offer more or less the same performance as standard desktop computers; they use the same software - the only technical difference being size, weight, and the ability to run on a battery. If operated mainly via a sensitive display (touch sensitive or electromagnetic), the device are also known as notepads or tablet PCs.

Mobile devices platforms

Mobile devices platform is the environment that supports development and operations of mobile

applications/services such as mobile messaging, e-commerce etc.
This includes:

- hardware/software environment for laptops, tablets smartphones and portable devices
- specifically designed environment that enables the functionality of mobile devices
- software on which other programs can run on mobile devices

Types Mobile devices platforms

i. Integrated Local platform
ii. Global platforms

Integrated Local platform

This is all services that support applications in the individual mobile devices
It consists of different mobile devices operating systems such as iOS, Symbian, Android, windows etc. and local system software services such as database managers, transactions managers and utilities for mobile devices.

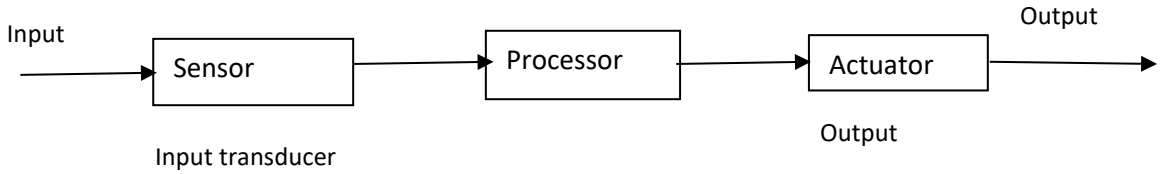
Global platforms

This is all Services that allows interaction of mobile devices with other portable devices in mobile ecosystem such social networks, Internet of Things (IoT) etc.
These includes:

- middleware services that interconnect mobile users, databases and application with each other.
- Network Services that transmit messages over wireless network.

Mobile Device Sensors and Tags

Sensor is a device or a module that can detect events or changes in its environment and share the information with other connected electronic devices. It can also be defined as an input device that receives and responds to a signal or stimulus. Sensors are transducers because they can convert signals from one physical form to another. Physical forms are mechanical, electrical, thermal, optical, magnetic etc. Sensor is an input transducer (e.g. Microphone) while actuator is an output transducer (e.g. Loudspeaker)



Sensors are essential components of smart objects, mostly small in size, have low cost and consume less power.

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Performance characteristics of sensors.

Input Range (Span): the minimum and maximum value of the physical variables that can be measured for instance -40F/100F in a thermometer. Output can be similarly defined.

Error

Error is the difference between the result of the measurement and the true value of the quantity being measured.

Error = measured value – true value

For example, measurement system gives a temperature reading of 50℃ , but the actual reading is 49 ℃, then the error is +1℃ (50℃ – 49℃). If the actual reading is 52 ℃ , then the error is -2℃ (50℃ – 52℃). The error can obtain in both positive and negative values.

Accuracy

It is the extent in which the value indicated by a measurement system might be wrong.

For example, if the temperature of the system have a specified accuracy of ± 5℃, this means that the reading given by the instrument to be lie within plus or minus 5 ℃ of the true value.

Accuracy is mainly expressed in percentage of the full range. For example, a transducer having an accuracy of ± 10% of full range output of 0 to 500 ℃, then the reading can be expected from plus or
minus 50 ℃ of the true reading i.e., from 450 ℃ to 550℃.

Sensitivity:

It is the relationship of output per unit input. For example, a resistance thermometer has sensitivity of 1Ω/℃. This shows that the thermometer having sensitivity, where there is a deflection
of 1Ω for every 1℃. This is also used to indicate the sensitivity to inputs other than being measured.

An ideal sensor will have a large and constant sensitivity. Errors associated with sensitivity are ‘saturation’ and ‘ddead-bands’

Linearity: Closeness of the calibration curve to a specified straight line (i.e theoretical behaviour least squares fit)

Monotonicity: A monotonic curve is the one in which dependent variable always increases or decreases ad independent variable increases

Hysteresis Error

Transducers can give different values of outputs to the same value of quantity being measured.

So the output value will be obtained by continuously increasing or continuously decreasing change. This effect is called hysteresis. The difference between the decrease in change and increase in change on the system of measurement known as hysteresis error.

Nowadays Sensors come integrated with many sensing elements and read-out circuitry in a single silicon chip, providing high accuracy and multiple functions.

Manufacturers use both advanced technologies and methods for signal processing and conversion

Features of latest sensors include: user-friendliness, accessibility and flexibility.

So there is a model shift in the sensor industry with integration of new technologies to make sensors
smarter and intelligent.

Ordinary sensors are still used in many applications but latest sensors based on innovation and advancement in microelectronics are becoming more and more intelligent, providing higher accuracy, flexibility and easy integration into mobile computing and distributed systems

An intelligent sensor structure

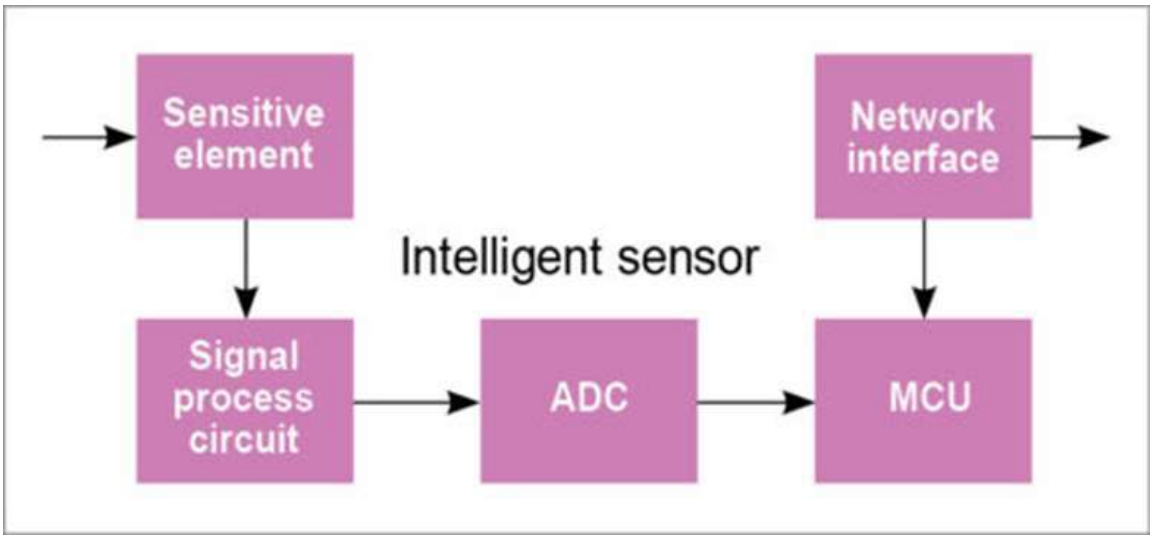

Figure 2.1: An intelligent sensor structure

An intelligent sensor may consist of a chain of analogue and digital blocks, each of which provides a specific function. Data processing and analogue-to-digital conversion (ADC) functionalities help improve sensor reliability and measurement accuracy.

Mobile devices based sensors

Smart device sensors have been embedded to enhance their usability, controllability, and management.

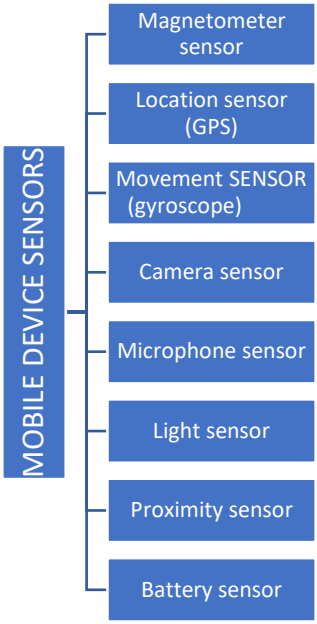

Figure 2.2: Mobile Devices based Sensors

Magnetometer: (digital compass): provides mobile phones with simple orientation in relation to earth's magnetic field.

- detects magnetic fields
- Can be used as a digital compass in applications to
- detect the presence of metals

Location Sensor also known GPS

- Detects location of phone which is one of the most important pieces of contextual information for smart applications.
- Location is detected using trilateral principle
- Distance is measured from three or more satellite (or mobile phone tower in the case of A-GPS) and coordinates are computed.

Movement Sensor (gyroscope)

- Detects orientation of phone
- Orientation measured using capacitive change when a seismic mass moves in a particular direction
  Gyroscope adds additional dimension to information provided by accelerometer for detection of phone orientation. Accelerometer measures linear acceleration of movement wvhile gyroscope measures the angular velocity.

Camera and Microphone Sensors

- Captures visual and audio information which can be analysed and processed to detect various types of contextual information. e.g. user's current environment and interaction can be inferred.
- Technologies like voice recognition and acoustic can be exploited for proper utilization of audio data.

Light Sensor

- Detects the intensity of ambient light
- Can be used for setting the brightness of screen and other applications in which there is action corresponding to the intensity of ambient light. E.g controlling the light in a room.

Proximity Sensor

The proximity sensor enhances the power management of the device; i.e., if it is near the user's ear, the screen will automatically turn off. This saves both battery life and accidental touch of the screen. Uses IR LED, which emit IR rays, the rays bounces back on striking an object.

- The distance can be calculated by difference in time, with this the distance of different objects from the phone can be measured.
- Can be used to determine when phone is close to the face while talking.

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The battery sensor controls the charging process and the temperature of the battery.

Some smartphones like Samsung's Galaxy S4 also have thermometer, barometer and humidity sensors to measure the temperature, atmospheric pressure and humidity respectively.

Smart phone sensors are being used to study different kinds of human behaviour and to improve quality of life.

Other sensors include:

- Medical sensors (worn on the body to monitor person's health)
- Wearable sensing devices available in the market that are capable of measuring different parameters such as heart rate, pulse, blood pressure, body temperature, respiration rate and blood glucose levels.

Sensor Materials

The exact sensor materials depend on their type and application. For example, digital, analogue, proximity and image sensors have their own materials, structures, fabrication techniques and packaging.

Types of Sensor materials

They are Passive and Active Materials (which categorized by roles played by the materials in the device, some play passive role, some play active role while some play both roles)

Passive Materials

- used to provide either mechanical structure or electrical connection. Some of these materials like silicon and gallium arsenide can also be used as active as well as passive materials.

Active materials

- essential to the sensing process in microelectronics, photosensitive, piezoelectric, magnetoresistive and chemoresistive films. Microsensor materials in the form of thin or thick films play an active role in the sensing system.

These devices are fabricated using chemical vapour deposition (CVD) or low-pressure chemical

vapour deposition (LPCVD) and special techniques like electrochemical deposition.

The scope and future Outlook of Sensors

- With micro and nano technologies, sensors can be used in automobiles and even human bodies.
- Use of intelligent sensors is also increasing in counter- terrorism, cargo tracking, biometrics among other applications.
- Latest sensors are used in automobiles to prevent impending crash and determine the type of airbags to be fired, and force and speed of their deployment.
- Increase in the use of MEMS (MicroElectroMechanical Systems) in medical applications, including implantable devices and handheld devices for diagnostics and monitoring systems.
- with advancements in technology, the new wave of sensors including IoTs and wearables is going to revolutionise the electronics industry in years to come.

Tags

TAGS are digital codes used for identification of devices, Smart tags seek out specific digital inputs and offer users convenient options.
Smart tags label data within documents or images when it identifies relevant words or numbers. It is like a digital version of a labeler, giving information about certain items it labels

Smart tags in mobile devices

In mobile tagging, the number of codes is essentially restricted to a dozen types. For reading out 2d-barcodes it is essential to install specific software,(a reader) on the mobile device.

- The reader uses the camera of a mobile phone for the mobile tagging process. The producers of the reader are very involved in developing solutions to increase the number of compatible mobile phones.
- Most services on the corresponding websites offer the download directly onto the mobile device or PC. In addition, many providers offer the user a generator for creating their own codes.

- Smartphone cameras can be used in Google's mobile Android operating system via both their
- own Google application or 3rd party mobile tag readers.

- Nokia's Symbian operating system features ascanner which can scan mobile tags, while m-barcode is a reader for the Maemo operating system.

- With BlackBerry devices, the App World application can natively scan mobile tags. Windows Phone 7.5 is able to scan tags through the Bing search app.

- In the Apple iOS, a mobile tagging reader is natively included in the iOS 11 update through the camera app, although more than fifty paid and free apps are available with both scanning capabilities and hard-linking to URI.

- Absence of a standard code makes a remarkable number of readers to be faced with

problem of inability to identify barcodes apart from their own proprietary codes.

- Though there are several readers which were created to read out non-proprietary code, these 'open source codes' are the most used for Optical Reading with Mobile like the QR Code and the Data Matrix.

Radio wave Frequency identifier (Rfid)

RFID is an identification technology in which RFID tag (a small chip with antenna) carries data which is read by RFID reader.
The tag transmits stored data through radio waves.

Characteristics

- Similar to bar code technology but does not require line of sight communication between the tag and the reader as is in the case of conventional bar code technology.
- It can identify itself without human intervention
- It not expensive and uses little power.
- Its range varies with frequency up to hundreds of meters.

Types of RFID

Active Tags	Passive tags
• Have power source	• Do not have power source, draws power from EM (electromagnetic) waves emitted by the reader
• Not cheap, shorter life	• Cheap and longer life

Types of RFID Technology

Near Technology	Far Technology
• RFID reader uses coil through alternating current is passed and magnetic field generated.	• RFID reader has a dipole antenna that propagates EM waves.
• The tag has a smaller coil which generates a potential due to ambient changes in magnetic field.	• Tag also a dipole antenna on which alternating potential difference appears and it is powered up.
• The tag produces small magnetic that encodes the signals to be transmitted, this can be picked up by the reader.	• Uses power to transmit messages.

Uses of RFID

- RFID is used to identify objects in IoT domain
- If RFID tags are integrated with sensing capabilities, they become sensors tags
- These Sensors' tags can sense data from environment, then the RFID reader can read this sensed data from the environment.
- Supply chain management
- Access control
- Identity authentication
- Object tracking

UNIT III:
SERVICE ORIENTED ARCHITECTURE (SOA)
SOA is defined as away to make software components reusable and interoperable through service interfaces. It is a paradigm for organizing and utilizing distributed capabilities that may be under the control of different ownership domains. It provides a uniform means to offer, discover, interact with and use capabilities to produce desired effects consistent with measurable preconditions and expectations.

Characteristics of SOA

- Services use common interface standards and an architectural pattern so they can be rapidly incorporated into new applications. This removes tasks from the application developer who previously redeveloped or duplicated existing functionality or had to know how to connect or provide interoperability with existing functions.

- Each service in an SOA embodies the code and data required to execute a complete, discrete business function (e.g. checking a customer's credit, calculating a monthly loan

payment, or processing

The Architecture

- Applications behind the service interface can be written in any other programming language, supplied as packaged software applications by a vendor (e.g., SAP), or obtained as open source applications. Service interfaces are frequently defined using Web Service Definition Language (WSDL) which is a standard tag structure based on xml (extensible markup language).
- The services are exposed using standard network protocols—such as SOAP (simple object access protocol)/HTTP or Restful HTTP (JSON/HTTP)—to send requests to read or change data.
- Service governance controls the lifecycle for development and at the appropriate stage the services are published in a registry that enables developers to quickly find them and reuse them to assemble new applications or business processes

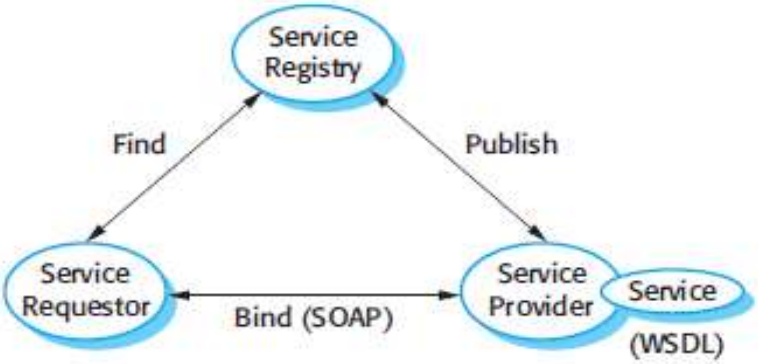

Figure 3.1: Service Oriented Architecture

Figure 3.1 summarizes the concept of a SOA. Service providers design and implement services and specify the interface to these services. They also publish information about these services in an accessible registry. Service requestors (service clients) who wish to make use of the service discover the specification of the service and locate the service provider. They can then bind their application to that specific service and communicate with it using standard service protocols.

Standards protocols for web SOAs are as follows:

- SOAP (Simple Object Access protocol)This is a message interchange standard that supports the communication between services. It defines the essential and optional components of messages passed between services.
- WSDL (Web Service Definition Language) is a standard for service interface definition. It sets out how the service operations (operation names, parameters, and their types) and service bindings should be defined.
- WS-BPEL This is a standard for a workflow language that is used to define process programs involving several different services.

Three Architectural Perspectives

Application Architecture

- business facing solution

- consumes services from one or more providers
- integrates them into the business processes

The Service Architecture

- a bridge between the implementations and the consuming
- applications.
- logical view of sets of services which are available for use.

The Component Architecture

- Various environments supporting:
- the implemented applications
- the business objects and their implementations

Benefits of SOA

- **Reusability:** The service oriented architectural approach supports scenarios for application integration, data integration, and service orchestration style automation of business processes or workflows.
- **Ability to leverage legacy functionality:** SOA enables developers to easily take functionality 'locked' in one computing platform or environment and extend it to new environments
- **Improved collaboration between business and IT:** In an SOA, services can be defined in business terms (e.g., 'generate insurance quote). This enables business analysts to work more effectively with developers on important insights.

UNIT-IV

FOUNDATIONS OF MOBILE AND UBIQUITOUS COMPUTING; MOBILE SOCIETY, INFORMATION SOCIETY, PERVASIVE AND SELF LEARNING ENVIRONMENT, INFORMATION AND PIRACY.

Mobile and ubiquitous computing is the paradigm that enables the integration of computing in our everyday life.

The Foundations of Mobile and Ubiquitous computing are

i. Mobile society
ii. Information society

Mobile society

The growing adoption of smart mobile devices, such as smart phones and tablets, is fundamentally changing the way business is conducted. New mobile technologies exert a significant influence on individuals, organizations, and society at large.

Mobile society can be said to be society in motion where you are not bound by your geographical location on things you can do or reach to.

Achievement of such society is one of the major motivating forces of this new innovation of mobile computing.

Information society

We are living in information age where the most valuable commodity is information which provides awareness, knowledge and power. Information Society can be defined as a society based on information and knowledge.

Characteristics of information society

Information society is global in principle, for geographic borders are not recognized by the flow of information.

Information society demands and promotes clarity, precision, honesty, and openness. Dishonest politicians and officials have no hiding place in this society.

Information society promotes equal opportunity. It has been a well-known fact since long time ago that "information is power." The free flow of information, in the information society, therefore, translates to equal distribution of power in this society. The availability of information to everyone without any restriction, control, or filtering, provides equal opportunity for all the citizens of information society

Pervasive and self learning environment information and piracy

This involves universal do it yourself environment where you can easily learn by surfing for information about any area of your interest.

This self learning gives opportunity for negative activities such as piracy, which is illegal and unauthorized usage or copying copyrighted software

UNIT-V

LOCATION TECHNIQUES AND SPACE MONITORING: POSITIONING TECHNOLOGIES, LOCATION SYSTEMS, LOCATION MANAGEMENT, POSITION FORECASTING, GEOGRAPHICS AND GEOMETRIC SPACE MODELS, SYMBOLIC SPACE MODELS

Location Techniques and Space Monitoring

Location system is concerned with the identification of devices' location at a particular time.

A wireless positioning location system consists of at least two hardware components;

- a measuring unit that usually carries the main part of the system and
- a signal transmitter which in its simplest case just sends a beacon signals.

Classification of Wireless Position Location

Classification based on the functionality of the hardware components and their interaction. A systematic classification of wireless position location systems can be categorized into three groups:

i. Handset-based positioning,

ii. Network-based positioning, and
iii. Hybrid-positioning system.

Handset-based Positioning (HBP)
In this system, the handset uses signals generated from multiple reference stations to calculate its own position. GPS is an example of handset-based positioning system.
Handsets with GPS receivers can estimate their position based on signals received from at least four satellites. The multiple transmitting satellites are in motion but their exact positions at the time of transmission is used by the node receiver to calculate its position.

Network-based Positioning (NBP)
Network-based location technology depends on current wireless networks to obtain information about user position. In these systems, several receiver stations measure the signals transmitted from wireless users and relay this information to a central site for processing to calculate the user's location. Examples of network-based positioning systems:

- cellular networks and
- Airborne Early Warning and Control System (AWACS)

The network-based system does not require changes of hardware or software in the user's node. Hence, can allow the positioning location of user's node in the operators system.

Hybrid-Positioning System (HPS)
A HPS creates positioning location estimate by merging both handset and network information' An example is the case where measurements are done by the handset and broadcasted to the network that executes the position location estimations.

Advantages

- The hybrid system is complex but could possibly provide superior accuracy,.
- Th Assisted GPS (A-GPS) is an example for the hybrid–positioning system.
- This system has the advantage of reducing the problems that occur in other positioning location
- systems by providing assistance information that enhance position estimates of nodes from one system for use by the other system.

The positioning location technique in GPS
The main goal of using GPS system is to find the position location of the user's devices. The principle behind GPS is the measurement of distance between satellites and the receiver device. The satellites send messages containing information about their exact orbits and the time when the message is sent beside other useful information. The receiver processes this information and measures time differences of these signals to obtain the position location.

The accurate point in positioning location methods can be calculated by using the basic principles of intersection of different lines and the proper geometric formulation to estimate the position location of a node. Several methods/techniques for determining the positioning location coordinates of the nodes include:

- Angle of arrival (AOA),
- Time of arrival (TOA),

- Time difference of arrival (TDOA),
- Received signal strength (RSS).

These methods built on geometric principles to calculate the position locations of a node using the lines and angles.
The basic mathematical principles used in position location are:

- Triangulation,
- Multilateration
- Hyperbolic

Triangulation principle

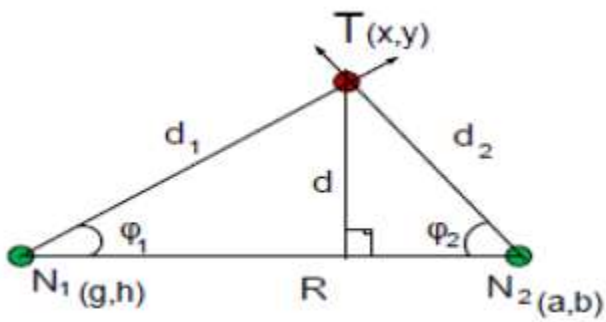

Figure 5.1: Triangular principle
uses the geometric properties of triangles to estimate a node positioning.
This principle consists of three points:

- two reference points (N1 and N2)
- a target point (T).

The intersection of line from N1 to T and from N2 to T, we get the two angles $\phi 1$ and $\phi 2$ as shown in the above Figure 5.1

The trigonometric functions are used to calculate the target position

$$R = \frac{d}{\tan \varphi_1} + \frac{d}{\tan \varphi_2}$$

$$d = \frac{R \,.\, \sin \varphi 1 \,.\, \sin \varphi 2}{\sin(\varphi_1 + \varphi_2)}$$

$$\varphi_1 = tan^{-1}\left(\frac{h-Y}{g-X}\right)$$

$$\varphi_2 = tan^{-1}\left(\frac{b-Y}{a-X}\right)$$

Where R is the line between reference point N1 and N2, d is the perpendicular line between target T and line R. d_1 is the line between target point T and the reference point N1, d_2 is the line between a target point T and reference point N2, $\phi 1$ is the angle between line d1 and line R. $\phi 2$ is the angle between d2 and line R, (g,h) are the coordinates of reference point N1 and (a,b) are coordinates of reference point N2, (XY) are the coordinates of the target T.

The coordinates of the target (X, Y) can be calculated as:

$$Y = x \tan \varphi_2 + (b - a\, tan\varphi_1)$$

$$X = \frac{b - h - a \tan \varphi_2 + g \tan \varphi_1}{\tan \varphi_1 - \tan \varphi_2}$$

The difference between reference points N1, N2 and target point T

$$d_1 = \|g - X\| = \sqrt{(g-X)^2 - (h-Y)^2}$$

$$d_2 = \|a - X\| = \sqrt{(a-X)^2 - (b-Y)^2}$$

Multilateration Principle

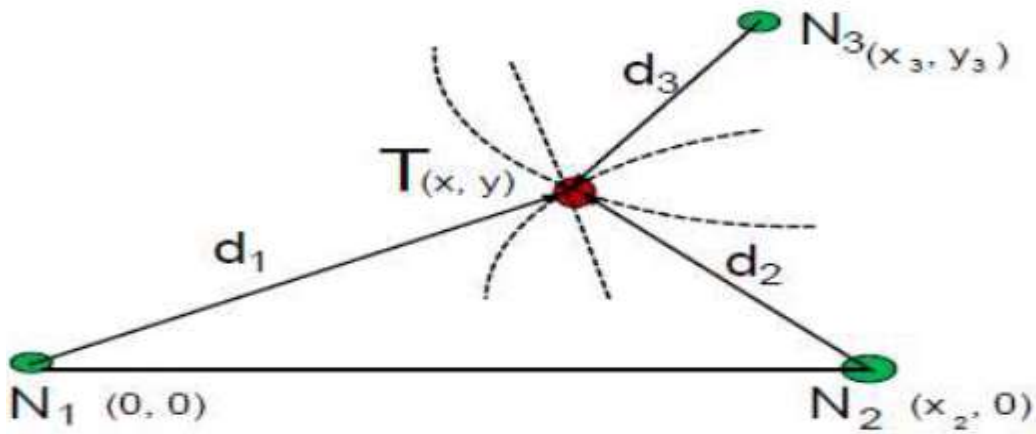

Figure 5.2: Multilateration principle

This is simply an extension of triangulation technique with more than three reference points.
• So, if there are three points N1, N2 , and N3, as shown in the above figure.
• The distances from the reference points to object T are known, then the point of the intersection of the three circles is the supposed location of object T.
• The first circle has radius (d1) and the second circle has radius (d2), the distances (d1 and d2) can be estimated by calculating the TOF (time of flight) of the signal.

To calculate the coordinates of the target T:
First, calculate the distances between reference nodes and the target T using the following equations:

$$d_1 = (t_1 - t_0) . c$$

$$d_2 = (t_2 - t_0) . c$$

$$d_3 = (t_3 - t_0) . c$$

Where c is the speed of light
T_0 is the time of a signal sent from T
d_1 is distance between N1 and T
d_2 is distance between N2and T
d_3 is distance between N3 and T
t_1 is arrival time of signal from T to N1
t_2 is arrival time of signal from T to N2
t_3 is arrival time of signal from T to N3

The equations for three intersecting circles with centers at the reference points and radii equal to distance from the target T are:

$$d_1^{\,2} = x^2 + y^2$$

$$d_2^{\,2} = (x - x_2)^{\,2} + y^2$$

$$d_3^{\,2} = (x - x_3)^{\,2} + (y - y_3)^2$$

Where d_1 is line between reference point N_1 and target point T
d_2 is line between reference point N_2 and target point T;
d_3 is line between reference point N_3 and target point T
The coordinates of the target point T are (x,y),
coordinates of reference point N_1 are (0,0)
coordinates of reference point N_2 are (x_2,0)
coordinates of reference point N_3 are (x_3,y_3).

Solving the three equations gives the coordinates:

$$x = \frac{x_2^2 + d_1^2 - d_2^2}{2\,.\,x_2}$$

$$y = \frac{x_3^2 + y_3^2 + d_1^2 - d_3^2 - 2\,.\,x\,.\,x_3}{2\,.\,y_3}$$

Hyperbolic principle

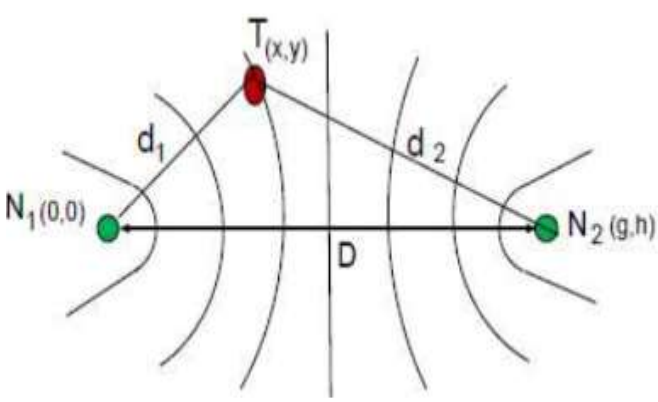

Figure 5.3: Hyperbolic principle

The hyperbola is a set of points in a plane whose distances to two fixed points in the plane have a constant difference.

• Each point represents a focus of the hyperbola which represents an anchor node or a reference point

The Positioning Location Applications

Positioning location technology becomes a critical attribute that can be used by everyone involved in the wireless value chain.

Areas of application include:

- Navigation services (science of finding the position of mobile objects and their way safely and efficiently to destination)
- Location Based Services (LBS)
- Climate and Environment Monitoring

Ubiquitous Computing

Ubiquitous Computing can be simply defined as global computing or digitally connected world. It is concerned with exploiting the increasing integration of services and (small/tiny) computing devices in our daily activities.

As have been mentioned earlier it requires particular solutions in many areas caused by dynamically changing computing environment such as users, devices and software components.

Examples: Internet of Things (IoT)

Internet of Things (IoT) is worldwide network of interconnected objects uniquely addressable based on standard communication protocols. It is new era of portable, wearable devices harnessing continuing advancements in mobile technology and sensor refinement.

Figure 5.4: Internet of Things (IoT)

Iot objects/devices are always referred to as “things”. It is a global infrastructure for the information society that enables advanced services by interconnecting physical and virtual things

Features of IoT

- contain embedded technology with unique identifier that enable them to connect to internet, smartphones and other IoT devices.
- The unique identifier allows each device or object to be individually identified and connected through its embedding computing system and interoperate within the existing

internet infrastructure.

- Can transfer and receiver data over the internet
- Things equipped with IoT technology can be configured, controlled or monitored using computer device such as smartphones.

Examples of IoT

- Security systems
- Thermostats
- Medical alert systems
- Farm equipment
- Light bulbs
- Cars
- Drones
- Speaker systems etc

Physical-Virtual Integration and Interaction models

Physical and software components are intertwined in IoTs, each operating in different spatial and temporal scales. They exhibit multiple and distinct behavioural modalities, interacting with each other in many different ways changing with context.
The IoT objects interaction with physical environment are enabled sensors and actuators.

Physical –Virtual Interaction Models include:

- Smart grid
- Autonomous automobile system
- Medical Monitory
- Process Control systems
- Robotics Systems
- Automatic pilot avionic

IoT Architecture

IoT has no one standard architecture, different architectures have been proposed by different researchers. These include:

- Three and five layer architecture
- Cloud and Fog based architecture
- Social IoT

Three and five layer architecture

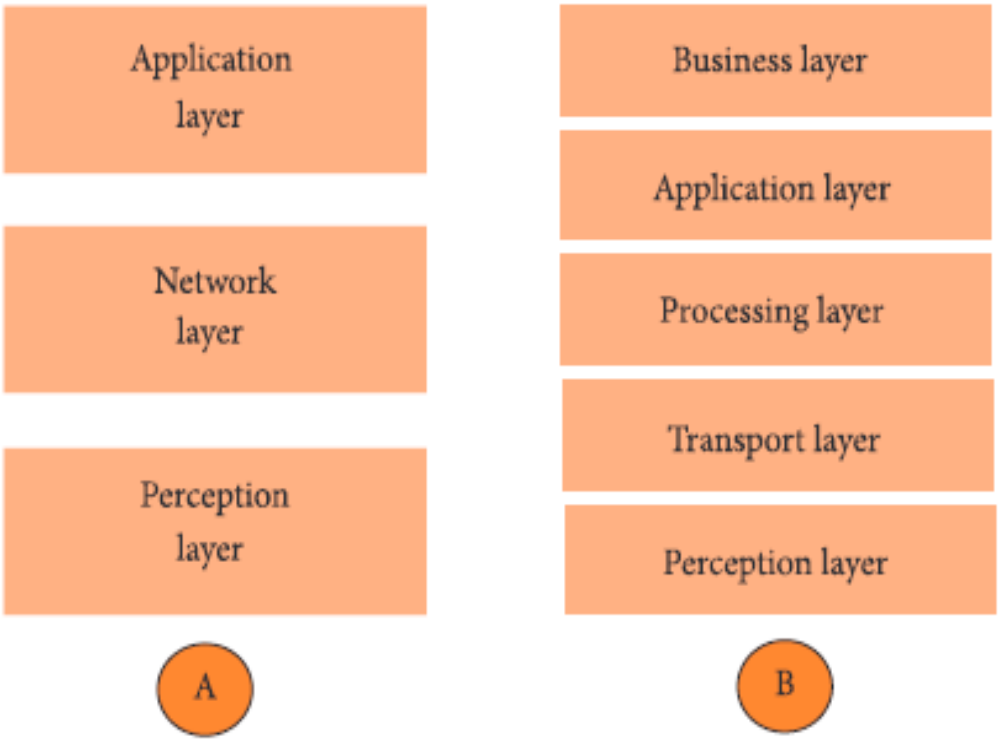

Figure 5.5: Architecture of IoT (A: three layers) (B: five layers)

The most basic IoT architecture is three layer which consists of:

- The Perception Layer
- The Network Layer
- The Application Layer

The Perception Layer

This is the physical layer which has sensor for sensing and gathering information about the environment, sensing physical environment or identifying other smart objects in the environment.

Network Layer

This connects other smart objects, network devices and servers, Its features are used for data processing and transmission.

The Application Layer.

This layer delivers application specific services to the users. It defines different applications on which IoT can be deployed such as smart homes, smart cities and smart health

Five layer architecture includes:

- Business layer

- Perception layer
- Processing layer
- Transport layer
- Application

Additional layers to three layer architecture are business and Processing layers. The role of perception and application layers are the same as in three layer architecture. The function of the remaining three layers are as follows:
The transport layer transfers the sensor data from perception layer to processing layer and vice versa through networks such as wireless, 3G/4G, LAN, Bluetooth, RFID and NFC.

The Processing layer also known as middleware stores, analyzes and processes large amount of data that comes from transport layer.It can manage and provide different set of services to lower layers employing many technologies like database, cloud computing and big data processing methods.
The Business layer manages the entire IoT system, including applications, business and profit models, and user's privacy.

Cloud and Fog Based Architecture
In Cloud Computing data processing is done in a large centralized fashion by cloud computers This architecture keeps cloud at the centre, application above it and network of smart things below
as is shown in Figure 5.6

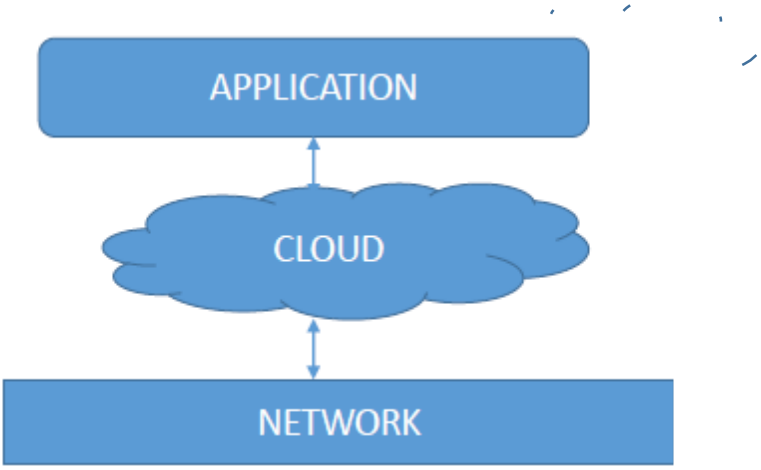

Figure 5.6: Cloud and Fog-based Architecture
Cloud computing is given primacy because it provides great flexibility and scalability. It offers services such as core infrastructure, platform, software and storage.

Fog Computing/Edge Computing
Fog computing and edge computing are interchangeably used though the latter predates the former. Fog computing refers to smart gateways and smart sensors while edge computing is slightly more penetrative in nature. It envisions adding smart preprocessing capabilities to physical devices like pumps, lights etc.
Fog/edge computing architecture presents layered approach which inserts monitoring,

preprocessing, storage and security layer between physical and transport layers. The monitoring layer monitors power, resources, responses and services. The preprocessing layer performs filtering, processing and analytics of sensor data. The temporary storage layer provides storage functionalities such as data replication, distribution and storage. Security layer performs encryption/decryption ensuring data integrity and privacy.

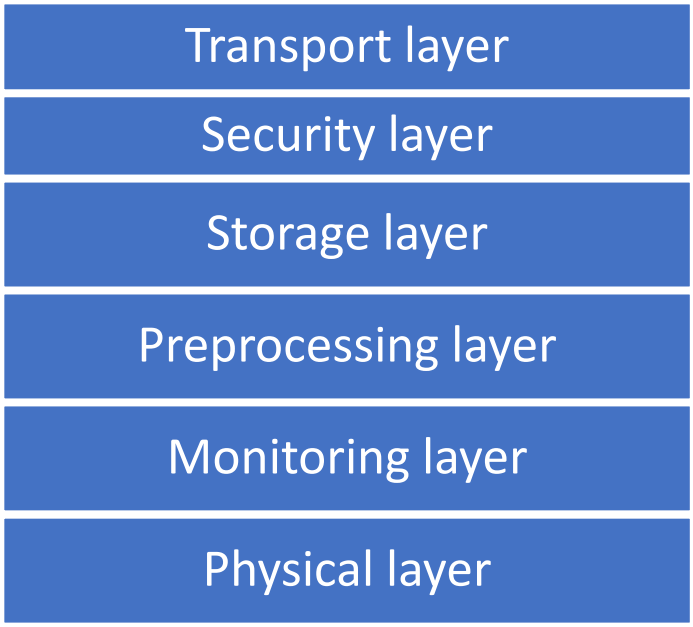

Figure 5.7: Fog Computing/Edge Computing

Social IoT
In a typical Social IoT (SIoT), devices and services are treated as bots were they can set up relationships between them and modify them over time. In this way, the devices seamlessly co-operate with each other to achieve a complex task.

The major components in SIoT model
ID: each object must be identified by assigned an ID to each based on traditional parameters such as MAC ID, IPv6 or any other identifying methods.

Metainformation: Along with ID, metainformation about the device that described its form and operation is needed. This is necessary to establish suitable relationships with the device and properly place it in the universe of IoT devices.

Security Control: similar to “friends list” in Facebook. A device owner may restrict the kinds of devices that can connect to the device. These restrictions are known as *Owner controls*

Service discovery: This is like a service cloud where there is a dedicated directories that store details of devices providing some kinds of services. This directories should be kept up to date such that devices can learn about other devices.

Relationship management: this module manages relationships with other devices. It stores the types of devices a given device can connect with based on the type of service provided. Eg. A light controller will be able to relate with a light sensor.

Service composition: This takes SIoT model to the new model to provide better integrated service to users. For example, if a user has a power sensor with his air conditioner and this device establishes relationship with an analytics engine, then it is possible for the ensemble to yield a lot of data about the usage pattern of the air conditioner.

Ubiquitous computing (UbiComp)Software architecture

Architecture of a system provides background for accurate system specification, which includes:

- Definition of components and their relationships
- Better understanding of system for proper implementation, maintenance and ongoing development.

Software architectures offers global and component view of the system.

Global view:
This focuses on the overall structure of software. It can take form of how the co-operating processes of the software interact and exchange messages.

Component-Based view:
This focuses on how the parts or components of the software are structured.

UbiComp software Concept

Device Operating System (DOS): drives the internal components of mobile device/object, and manages other applications

Middleware: Ubiquitous computing is the core of the Internet of Things, which means incorporating computing and connectivity in all the things around us. Interoperability of such heterogeneous devices needs well-defined standards. But standardization is difficult because of the varied requirements of different applications and devices. For such heterogeneous applications, the solution is to have a middleware platform, which will abstract the details of the things for applications thereby facilitates the interaction between smart objects in smart space. The middleware abstracts the hardware and provides an Application Programming Interface (API) for communication, data management, computation, security, and privacy. Note that main requirements of middleware are interoperability and abstraction.

The challenges, which are addressed by any IoT middleware, are as follows:

- *Interoperability and programming abstractions*: for facilitating collaboration and information exchange between heterogeneous devices, different types of things can interact with each other easily with the help of middleware services.
- *Device discovery and management:* this feature enables the devices to be aware of all other devices in the neighborhood and the services provided by them.

- *Scalability*: a large number of devices are expected to communicate in an IoT setup. Moreover, IoT applications need to scale due to ever increasing requirements. This should be managed by the middleware by making required changes when the infrastructure scales.
- *Big data and analytics:* IoT sensors typically collect a huge amount of data. It is necessary to analyze all of this data in great detail using some sophisticated machine learning algorithm.
- *Security and privacy*: The middleware should have built in mechanisms to address issues of security and privacy, along with user authentication, and the implementation of access control.
- *Cloud services:* the cloud is an important part of an IoT deployment. Most of the sensor data is analyzed and stored in a centralized cloud. It is necessary for IoT middleware to seamlessly run on different types of clouds and to enable users to leverage the cloud to get better insights from the data collected by the sensors.
- *Context detection:* the data collected from the sensors needs to be used to extract the context by applying various types of algorithms. The context can subsequently be used for providing sophisticated services to user.

Examples of middleware are:

- Oracle's Fusion Middleware
- OpenIoT middleware
- Hydra middleware
- FiWare

Classification of Middleware

Middleware can be classified based on their design as follows:

i. Event based:
ii. Service oriented
iii. Database oriented
iv. Semantic
v. Application specific

Event based: here, all the components interact with each other through events which generated by producers and received by the consumers. Each event has a type and some parameters. This can be viewed as a publish/subscribe architecture, where entities can subscribe for some event types and get notified for those events.

Service oriented: these are based on Service Oriented Architectures (SOA), in which independent modules provide services through accessible interfaces. A service-oriented middleware views resources as service providers. It abstracts the underlying resources through a set of services that are used by applications. The services are published by providers in a service repository from where the consumers can discover services and then bind with the provider to access the service. Service oriented middleware must have runtime support for advertising services by providers and support for discovering and using services by consumers. Example of this class is Hydra.

Database oriented: in this approach, the network of IoT devices is considered as a virtual relational database system. The database can then be queried by the applications using a query language. There are easy to use interfaces for extracting data from the database. This approach has issues with scaling because of its centralized model. Semantic: semantic middleware focuses on the interoperation of different types of devices, which communicate using

different formats of data. It incorporates devices with different data formats and ontologies and ties all of them together in a common framework. The framework is used for exchanging data between varied types of devices. For a common semantic format, we need to have N adapters for communication between N devices because; for each device, we need N adapters to map N standards to one abstract standard. There is a semantic layer is in which each resource is mapped to a software layer for that resource. The software layers then communicate with each other using a mutually intelligible language (based on the semantic web). This technique allows multiple physical resources to communicate even though they do not implement or understand the same protocols.

Application specific: this type of middleware is specific purpose solution. It used specifically for an application domain for which it is developed because its whole architecture is fine- tuned on the basis of requirements of the application. The application and middleware are tightly coupled.

Common IoT middleware are:

.F*iWare***:** contains a large body of code, reusable modules, and APIs that have been contributed by thousands of FiWare developers. Any application developer can take a subset of these components and build his/her IoT application.

OpenIoT: an open source initiative with 7 different components.

Physical plane: This is at lowest level and collects data from IoT devices and also does some preprocessing of data. It has different APIs to interface with different types of physical nodes and get information from them.

Virtualized plane: This is the next to physical plane. It has 3 components

i. The scheduler
ii. The data Storage component
iii. Service delivery

The scheduler: manages the streams of data generated by devices. It primarily assigns them to resources and takes care of their QoS (Quality of service) requirements.
The data storage component: manages the storage and archival of data streams.
Service delivery component: processes the streams. It has several roles. It combines data streams, preprocesses them, and tracks some statistics associated with these streams such as the number of unique requests or the size of each request.

The application layer: This is the uppermost layer with 3 components:

i. request definition,
ii. request presentation,
iii. configuration.

The request definition component: helps us create requests to be sent to the IoT sensors and storage layers. It can be used to fetch and query data.

The request presentation component: creates mashups of data by issuing different queries to the storage layer.
The configuration component: finally, helps us configure the IoT devices.

Network Operating System (NOS): facilitates transmission of data/information via network

Applications

Wireless (Mobile) Application: uses three tier architecture

i. Presentation tier
ii. Logic tier
iii. Data tier

Presentation tier

- User interface is the top most level of App.
- Main function of App is to translate task and results to something the user can understand.

Logic Tier

- This layer coordinates the application processes commands
- Makes logical decisions, evaluations and performs calculations
- Moves and processes data between the two surrounding layers (Presentation and Data layers)

Data Tier

- At this layer, the information is stored to and retrieved from database or file system,
- Then the information is then passed to Logic tier for processing and then eventually to user.

For Web Application

Presentation Layer: Consists of the browser and dynamically generated HTML

Business Layer: consists middleware files (function-specific PHP, Java servlets); Web services and related functionalities

Data Layer: consist of database server using Java Servlet; module that talks with database

It intertwines with Business layer to a degree.

Importance Three Tiers Architecture

- Keep the presentation separate so it's lightweight, easier to maintain, and can be tested separately
- Keep the logic separate so you can change the logic as needed without having to change the presentation too much.
- Keep the data separate because you should NEVER build a system based on the current data values

Design approaches for each layer

Presentation Approach

- Remember your client type
- Determine how you will present data in a coherent,unified method
- Determine how you will guard against untrusted input
- Ensure you have factored out your business logic
- Determine how you will pass data between layers (i.e. how you will call the service, how you will get more info about a building, etc)

Business Approach

- Identify FEATURES that will exist at this level
- Build components that support a feature's execution
- Hide implementation details from the presentation layer
- Determine if (how) you will cache information on the device
- Map out use cases

Data Approach

- For each feature, determine what data is required
- Build SQL queries around the features
- Ensure that you are using prepared statements to guard against incorrect data entry (or injection)
- Determine how you will manage connections
- Determine if you will batch up commands into one big command

Design Principles For Smart Spaces Pervasive Computing

Space

- In geographical theory, space refers to abstract geometrical extension and location, while place describes our experience of being in the world and attaching meaning, memories and feelings to physical location.
- Physical locations have purpose, provide opportunities for specific circumstances to arise, have values and history.

Many pervasive systems today use location as a substitute for domain and a domain has embedded understanding and protocols as to what is regarded as appropriate behaviour.
Unfortunately, the notion of location lacks many important qualities, such as the social dimensions, that are intrinsic to the notion of a domain.

Categories of Spaces

Spaces can be categorized into:
Private space (excludes one from the public, creates opportunity for privacy)
Social space (provides interaction for group of people within public space)
Public space (accessible to everybody with equal rights based on the guiding rules places by the owner)
Private, Social and Public locations are defined not by geographical coordinates but possible actions they allow.

Smart Space

Smart Space differs from spaces provided by physical environment. Spaces provided by computing devices and other smart objects can be regarded as interaction space.
Interaction spaces are means by which digital technology can access the information spheres.
They depend on the type of technology used, as well as the physical characteristics and affordances of the technology. Similar to spaces defined by the physical environment, interaction spaces may be private, social or public.

Example of Interaction Space:

The plasma screen positioned in front of two people creates a social interaction space that

includes both of them.
The person on headphones though within the public space, the headphone creates a private interaction space for him.
Another person with PDA can create different types of interaction spaces, depending on its position and orientation.

Private Spaces

- promote a sense of security and privacy.
- Artefacts/objects that are usable by only one person create private interaction spaces.
- Headphones for instance create a private interaction space, even if the person who wears them is in a public space.

Social space

Groups of people interacting collaboratively with objects are usually in a social interaction space. E.g. The interaction space that is created by the board game is a social interaction space.

Public space

- An artefact/object that makes a resource available to everyone present in a public space generates a public interaction space. E.g. a large screen showing a football match in a public square creates a public interaction space.

Characteristics of Pervasive Computing System

A pervasive computing system available in a public, social or private space should promote the following characteristics in order to function in harmony with the space itself:

- Variety (the range of users and activities)
- Robustness(the ability to change the use of space),
- Visual appropriateness (the degree to which the visual cues make it clear what options are available),
- Richness (the range of sensory experiences like smell, lighting and touch),
- Personalization (the ability of inhabitants to customize their environment)

Design of Pervasive Computing

A pervasive system may be viewed as:

- Set of digital objects working together. Like their physical counterparts, these digital objects have a huge impact on the way a space is used and perceived, and on the results and effects it has on people.
- In designing these objects, either digital or physical, we must expect their effects and try to tailor and direct them according to our goals.
- Invisible part of /or extension to the physical environment must:
    - ➢ encompass our aspirations and goals both when viewed as part of the physical environment and when considered on its own.

High level approach to design of pervasive computing system involves:

- Effective integration of spaces which is manipulated by architecture and interaction space which is manipulated by computer systems.

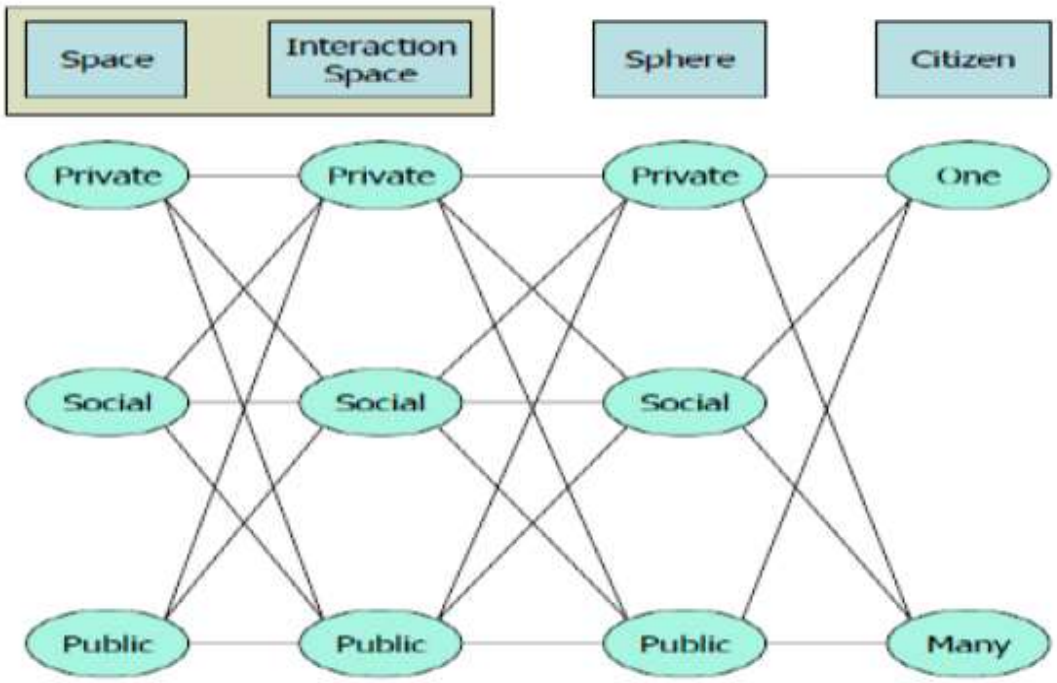

Figure 5.6: High level approach to design of pervasive computing

Key characteristics of Public Pervasive Computing System Design

- They are accessible
- They allow people to engage in activities
- They are comfortable
- They are sociable

Issues to be considered when designing pervasive system based on these characteristics.
Accessibility:

- the presence of the pervasive system should be made visible
- The system should be easily recognized and identified
- There should be easy connectivity, accessing and usability of the system

Comfortability:

- They should augment comfort in public space
- Any sensory for instance visual and auditory manifestation of pervasive system should be attractive to owners and users of public space.

Security
It should enhance safety and security in the public space.

Real world deployment of pervasive system.

- System deployment is the process between the production and the execution of software systems,

- It involves a set of correlated activities that consists of making configurations and bringing the software to its desired execution state.
- This process can continue along the lifetime of the software system in order to bring it to a new state via reconfigurations and updates

Implementation

- One of the most difficult tasks in implementing a pervasive computing application is possibly how to cope with the heterogeneity of the devices involved.
- Sometimes teams need to employ knowledge in microcontroller programming (assembler, C), mobile device (restricted C++ or Java), and platform independent software development including web-based services at the same time
- They also have to tackle other issues of distributed and mobile systems like various data types, interrupted communications, and roaming users

Deployment

- Deploying an application can happen in various levels and stages.
- The most general situation is that a system that is made available for public access, will work independently and run unsupervised.
- Some social, legal, as well as economic issues make deployment of pervasive applications difficult even if most standard problems concerned with distributed, ad-hoc
  systems such as name resolution, configuration or error management have already been taken care of.

Evaluation

- A major part of the development cycle, especially when following a user-centred design process is governed by evaluations.
- Most of the different types of prototypes imply specific types of evaluations
- Different methods of evaluating a system have been introduced and proven valuable, depending on the specific stage in the development process, the desired results and the amount of time (and money) that is available.
- Some of the considerable factors are user attention, device invisibility, and application robustness.